Consolidated draft functional requirements

No	FR (draft)
FR II-1	Escape routes shall allow all persons to reach timely a position of safe evacuation starting from the position normally occupied regardless of the emergency, its impact on the vessel, ship's conditions, and environmental conditions. Expected Performances: • At least two mutually independent routes • Each escape route shall have sufficient capacity to facilitate escape of the anticipated maximum number of persons from the position to the embarkation position regardless of their capabilities (disability or impairment) • Escape routes shall be usable under anticipated emergency conditions of the ship (list, trim, movement, light) • Escape routes shall be identifiable for all persons under all conditions • Minimize risk of injury when using escape route • Timely: *see NOTE 1* • Minimize the impact of the emergency (fire, explosion, water ingress, structural collapse, etc) on the availability of escape route • Protected from objects that hinder or prevent escape • Consider crowd dynamic/effects on the escape (speed, congestion, panic, freeze • …
FR II-2	All persons onboard shall be provided with instruction and familiarisation regarding the escape routes from the position normally occupied. Expected Performances: • Familiarisation at the time of embarkation, including the needs of disabled/injured person • Periodic training, including the transporting of disabled/injured person • Periodic drill of evacuation shall include the transporting of disabled/injured person • Instruction and information on the escape route, easily accessible by all persons at all times Additional aspects to be addressed: • Minimum requirements for the necessary specification for instruction and familiarisation
FR II-3	All persons on board shall be guided to the position of safe evacuation Expected Performances: • Crowd management system to ensure that all persons reach the position of safe evacuation starting from the position normally occupied.
FR III-3	Survival craft shall be capable of being timely embarked by all assigned persons, taking into account anticipated adverse weather conditions Expected Performances: • Shall be timely embarked in adverse weather conditions (in relation to area of operation), i.e. up to *see NOTE 2* - Wind - Wave - Wave spectrum peak period - Temperature a) Air: b) Water - Icing

	- Green water - Day and night - Poor visibility • Survival craft boarding arrangement, and the ship are compatible • Embarkation shall be free of obstacles and/or obstructions • Launched survival craft shall be accessible from the ship for all assigned person • Empty floating survival craft shall be accessible by an able-bodied person from the ship or the water without assistance • Embarkation shall be adequate for all assigned persons, including non-able-bodied persons with necessary equipment • Embarkation shall be adequate for all assigned persons with necessary PPE
FR III-5	Timely abandonment shall be possible for all persons on board regardless of the emergency Expected Performances: • The evacuation/abandonment process shall address emergencies that challenge successful evacuation according to FR II-1 (proceed to embarkation) (imminent danger of capsizing and foundering) • Without intervention, survival craft shall be available and usable even after the ship is lost • Means to verify which persons have abandoned the ship • The ship shall provide means to safely reach the water
FR IV-1	All survival craft systems shall be ready for safe use, immediate launch and move to a safe distance from the ship regardless of the emergency, its impact on the vessel, ship's conditions, and environmental conditions. Expected Performances: • Equipment is operational and usable for all static conditions or the ship (*see Note 3*) • Adverse weather conditions (in relation to area of operation), i.e. up to *see NOTE 2*
FR IV-2	All survival craft shall be able to safely and timely reach a safe distance from the ship and maintain it until retrieval Expected Performances: • Adverse weather conditions (in relation to area of operation), (see Note 2) • Passive or active manoeuvrability for safe operation • Stable in manoeuvring abilities (for propelled ~ Res. 137(76) for basic principle), easy to navigate (prevent CT, CN and GR); - Even after loss of propulsion - Including all unpropelled - Maintain propulsion and steering after single impact • Self-propelled: means to mitigate risk of GR
FR IV-3	All survival craft (systems) shall operate in such a way that the risk of accidents or delays is minimised Expected Performances: • Minimise the risk of "faulty" operation - General: a) Common communication basis (IMO Standard Marine Communication Phrases) b) One failure tolerant (human and/or machine) c) Protect against third party intervention (including cybersecurity?) d) Human – machine interface, human centred design

	- If personnel assigned/involved a) Training and drill with the survival craft systems installed on board • Crowd behaviour: panic • Minimize possibility of interference between SC
FR IV-4	The total capacity of the survival craft shall be sufficient to accommodate all persons on board, taking into account any loss of capacity in the survival craft due to the emergency or failure in the survival craft system Expected Performances: • Total capacity shall include spare capacity to accommodate for maximum anticipated loss of capacity due to but not limited to: - FX: pax: 1 MVZ both sides; cargo: probabilistic model to offer the whole design space - CN: probabilistic model based on damage stability damage extend - CT - GR: probabilistic GR model (reference is made to EU funded research project FLARE) - blocked launching area, the failure of any survival craft system shall not prevent the launching of any other survival craft - 'malfunction' / inoperable SC capacity
FR IV-5	All survival craft shall protect all persons on board until rescue Expected Performances: • Survival craft system shall be seaworthy providing: - structural integrity over design life - sufficient strength to maintain its functionality under all anticipated operational loads - protection from lightning - protection against occupant injury caused by contact with structure and/or fitting (for anticipated adverse sea/weather conditions) including launching - ability to withstand collision with ship's side during launching (Fully loaded survival craft, 3.5 m/s impact test for davit launched survival craft (81(70) para 6.4.1) - ability to withstand collision between SC CT - a safe place after single impact penetrating hull or buoyancy chamber (positive stability, no flooding or water ingress of occupant space) - protection against water ingress - sufficient propulsion and steering after single impact - buoyancy for all loading conditions taking into account anticipated weight of persons on board (e.g. mean + Stdv) - rolling period ≥ [8 s to 10 s] (see also habitable environment). - protection against negative interaction with marine life in the area of operation • early detection of fire, toxic gases, and means for suppression/extinguishing of any fire
FR Va-1	Each ship shall have means to provide habitable conditions for all persons on board until retrieval after abandonment of the ship in distress Expected Performances: (*for habitable environment see Note 4*) • Survive for the maximum anticipated time for retrieval/rescue • Body core temperature shall be maintained for survival • Consider wet and dry clothing • Protect against rain, sun

	• Means to prevent hypothermia, hyperthermia • Atmosphere (CO_2, toxic …) • Behaviour in a seaway / sea sickness • Safe seating • food/water rations shall be provided for all persons in the survival craft for the anticipated time until retrieval • Daily supply of food shall be sufficient to maintain vital function (calories/Joule) • Additional supply for persons with special needs • Daily water supply shall be sufficient to maintain vital function (~1.25 l/day) • Additional daily water supply for persons with special needs • Minimise probability and consequences of food poisoning (allergy, medicine) • Means to deal with waste products and sanitation
FR IV-5	Each survival craft shall provide for elementary first aid Expected Performances: Tbc.
FR Vb-1	Medicine required by persons with special needs shall be made available Expected Performances: Tbc.
FR Vb-2	Promote positive human behaviour supporting survival from deployment to time of rescue Expected Performances: • Familiarisation with crisis management • Periodic training • Periodic drill Additional aspects to be addressed: • STCW Section A-V/2, V/3
FR VI-1	All survival craft shall be detectable under all conditions Expected Performances: • Shall be readily detectable under adverse weather conditions and regardless of visibility • Detectable from air and water surface
FR VI-2	Means independent of the ship shall be provided to communicate on the emergency Expected Performances: • Means shall be provided to exchange information with coastal state/institution in charge • • Send and receive • Training • Drill • Information shall be available for SAR on - Location - Number of SC - total number of person on board of the ship
FR VI-3	All survival craft shall be able to remain in close proximity to each other Expected Performances:

	• Training • Drill • Means for connecting survival craft
FR VI-4	All VoO shall be equipped with efficient means for rapid retrieval of survivors from the survival craft Expected Performances: • ship-specific plans and procedures for retrieval of persons • specific equipment intended to be used for retrieval purposes • specific equipment adequate for anticipated adverse weather conditions • minimize the risk to shipboard personnel involved in retrieval operation • consider disabled, injured and impaired persons • training and drill
FR VI-5	All survival craft shall allow for safe and timely transfer to means used for retrieval Expected Performances: • all means of rescue (VoO, helicopter)

Notes used in the consolidated functional requirements

NOTE 1 Timely

The term “timely” means that all persons can abandon the ship before the ship capsizes or sinks, for example, and thus no longer allows for further evacuation. The close relation between the conditions of the ship in distress and successful abandonment had been highlighted in the EU funded research project FLARE[1,2], i.e. the relation between “time to capsize” (TTC) and “time to evacuate” (TTE). Since TTC depends on the damage and weather conditions, it is obvious that “timely” is not a fixed time.

Based on IMO requirements on evacuation [4], the total evacuation time *n* in ***passenger ships***, is estimated as T_{tot} with the following performance standard and using a specific parameter contained in the mentioned IMO requirement:

$\mathbf{1,25}$	$(\boldsymbol{R}+\boldsymbol{T})$	$+\frac{\mathbf{2}}{\mathbf{3}}$	$(\boldsymbol{E}+\boldsymbol{L})$	$\leq \boldsymbol{n}\ (\mathbf{1})$
(/Safety factor	*Response duration:* *10 min night* *5 min day* *Travel time depends on various factors*	*Overlap duration: 1/3 (E+L)*	*Embarkation and launching: ≤ 30 min (pax)*	*Ro-ro pax: 60 min* *Cruise: 60 min ≤ 3 MVZ* *Cruise: 80 min > 3 MVZ*

By this *simplified method*, the total evacuation duration is calculated and compared with an allowable time n, which depends on the vessel type (ro-ro or other passenger ship), and the number of Main Vertical Zones (MVZs). This represents the time necessary for all persons on board (passengers and crew) to reach the pertinent assembly stations, embark on the survival crafts and launch them.

[1] Hamann R., Eliopoulou E., Zaraphonitis G. (2019): Revised event tree and recommendations for improvements. FLARE D2.5.

[2] Vassalos D, Luhmann H, Cardinale M, Hamann R, Papanikolaou A, Paterson D. (2022) A Multi-Level Approach to Flooding Risk Estimation of Passenger Ships. SNAME Conference Houston.

The real evacuation time, for passenger vessels, depends on various factors such as those contained in the below Figure 1 – (taken from the research study “A Bayesian network model for evacuation time analysis during a ship fire”) not only linked to the dimensions of the escape route and means of escape.

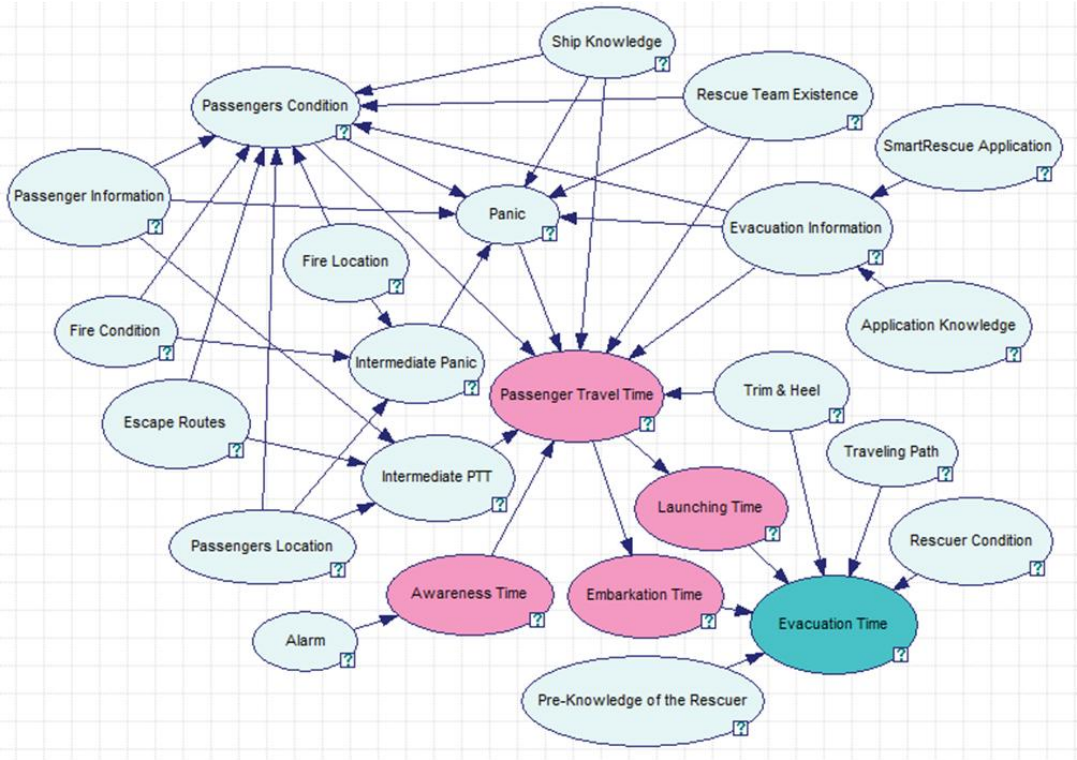


Figure 1 The BN evacuation model

Factors influencing the evacuation time, the estimated time to evacuate (TTE):

- Ship’s geometrical factors
  - Relatively narrow passageways
  - Length of escape routes
  - Staircases
  - Number of decks to be overcome
- Human effect (uneven evacuation efficiency)
  - passengers, as temporary residents onboard, are more likely to choose routes they are familiar with or follow the crowd
  - common routes connecting public and living areas may be crowded
  - physiological characteristics, e.g. shoulder width
- Ship’s environment (list and trim) and population and procedures
  - impact of heeling and trimming on human movement. Figure 2 shows the effect of different heeling angles on the person flow, i.e. the walking speed. As shown angles above 10° led to a significant reduction in walking speed.

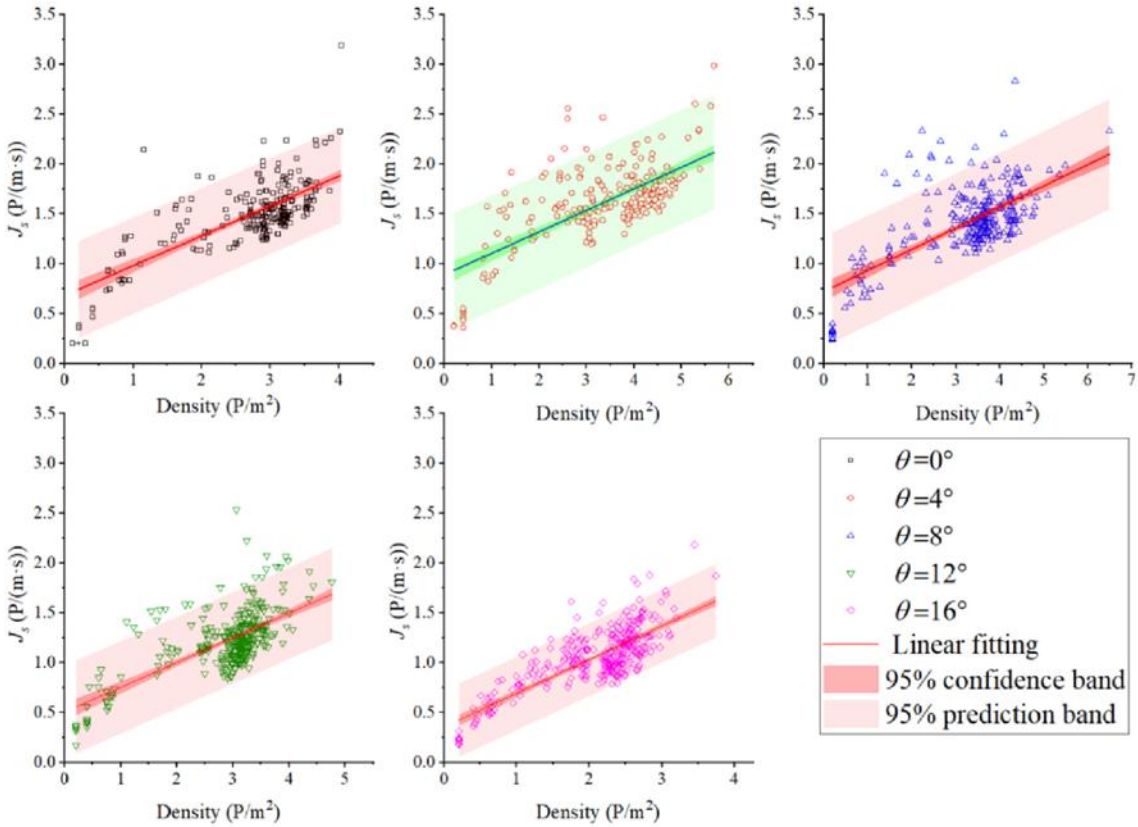


Figure 2 Density-flow rate (J) fundamental diagrams at different heeling angles (plotted vs. person density)[3]

Further work is needed to make recommendations on “timely”.

[3] *Experimental study on human evacuation onboard passenger ships considering heeling angle and opposite directions* (Siming Fang et all – June 2024)

NOTE 2 Adverse weather conditions

The following parameters characterise weather conditions relevant for life-saving appliances:

- Wind
- Wave
- Wave spectrum
- Temperature
- Humidity

It is recommended to determine the limits up to which the equipment shall be operable by means of representative weather data and agreed statistical thresholds (percentiles). For instance, using the North Atlantic data and 0.1% probability of exceedance, yields a wind speed of 20 m/s and 8 m significant wave height.

Temperature:

- Air:
  - specify extreme values temperature based on weather data for last 20 years and use mean ± 2Stdv otherwise depending on operational profile/area,
  - LSA Code operability -15°C to +40°C, stowage -30°C to + 65°C
- Water:
  - specify extreme temperature based on weather data for last 20 years and use mean ± 2Stdv otherwise depending on operational profile/area,
  - LSA Code operability -1°C to +30°C, stowage -30°C to + 65°C)

NOTE 3 Ship static conditions

The conditions of the ship in distress are crucial for the usability and operability of life-saving appliances and escape routes. Relevant parameters are list and trim of the ship. Both parameters can be used to specify the operational and usability envelope for life-saving appliances. For instance, it is useless to require operability of the life-saving appliances for list angles outside the ship's stability as the ship will capsize. Besides the operational limits for life-saving appliances the reconciliation of all elements used in the escape and abandonment process is recommended.

Intact and damage conditions vary depending on ship's size, type, design, construction, etc., and it may not be possible or meaningful to determine trends or certain value of ship's boundary conditions for list and trim. However, it is obvious that the ship's stability can be used to determine operating limits for LSA.

It should be noted, as existing ships are constructed based on the requirements in various IMO Instruments, finding the boundary limit within the IMO requirements may be a way to find certain values for the excessive condition of ship's trim and list.

For determining the ship's damage conditions, the following ship type dependent IMO instruments are available[4].

Table 1 Applicable Damage Stability Requirements for Each Ship Type

Ship type		**Applicable damage stability requirements**
Passenger ships		SOLAS II-1/6 to 8
Cargo ships having a length of 80 m and upwards		SOLAS II-1/6 to 7-3 or ICLL Reg.27 as appropriate
Oil tankers	Type "A" ship with assigned freeboard less than type "B" having a length of more than 150 m	MAROL Annex I/Reg.28 and ICLL Reg.27
	Other than the above	MARPOL Annex I/Reg.28
Gas carriers	Type "A" ship with assigned freeboard less than type "B" having a length of more than 150 m	The IGC Code Chapter 2 and ICLL Reg.27
	Other than the above	The IGC Code Chapter 2
Chemical tankers	Type "A" ship with assigned freeboard less than type "B" having a length of more than 150 m	The IBC Code Chapter 2 and ICLL Reg.27
	Other than the above	The IBC Code Chapter 2

Damage stability requirements in various IMO instruments suggest its boundary list conditions as in Table 2 below. In this regard, TRange or residual stability ranges address ship's heeling movement at ship's equilibrium after damage (i.e. dynamic stability condition). These values

[4] It is noted that these instruments are used for providing an adequate safety of a ship after water ingress

are the heeling limit set out in IMO instruments, thus, ship's actual worst list condition after damage may be smaller than as listed below. It is also noted that the value of extreme list may be unrealistic to apply to evacuations and LSA.

Table 2 Extreme Ship's List in IMO Instruments

Regulation	**Equilibrium**	**Maximum heel angle after damage**	**TRange or Residual Stability Range**	**Extreme list as allowed by IMO instruments**
SOLAS regulation II-1/7.3	All flooding stages	15° for passenger ships; and 30° for cargo ships.	20° for ro-ro passenger ships; and 16° for others	35° for ro-ro passenger ships, 31° for passenger ships (other than ROPAX); and 46° for cargo ships
ICLL Reg.27(13)(c) and (e)	Final waterline	15° (or, if no part of the deck is immersed, 17°)	20°	35° (or, if no part of the deck is immersed, 37°)
MARPOL Annex I/Reg.28.3.2 and .3	Final waterline	25° (or, if no deck edge immersion occurs, 30°)	20°	45° (or, if no deck edge immersion occurs, 50°)
IGC Code, paragraphs 2.7.1.2 and .3	All flooding stages	30°	20°	50°
IBC Code	All flooding stages	25° (or, if no deck immersion occurs, 30°)	20°	45°

Regrettably, boundary conditions of ship's trim, from a point of damage stability, are not defined in IMO instruments. Therefore, such conditions will need to be assessed for each individual ship.

Based on the above it is recommended that the expected performance makes reference to the individual limits of a ship for list and trim determined on the basis of the damage stability calculation using either the IMO models or other accepted model.

NOTE 4 Habitable environment

The wording habitable environment is used in IMO instruments, e.g. SOLAS Chapter II-1 and the Polar Code. The Polar Code defines habitable environment as a "ventilated environment that will protect against hypothermia". As shown in F1, atmosphere and temperature are two important factors, but noise, provisions and space should also be considered.

In the following initial information to be used for characterising a habitable environment, i.e. specifying thresholds, are summarised. It should be noted that this list is not complete as the importance of other aspect health aspects need further evaluation, e.g. hygiene, musculoskeletal and circulatory health.

Atmosphere

The term atmosphere is usually associated with O_2 and CO_2 concentrations. However, other gases may also be relevant, for instance when alternative fuels are used for propulsion, and therefore the list was amended by other gases.

Chemical	**Maximum allowable concentrations**			**Source**
	1 hr	8 hrs	24 hrs	
	ppm	ppm	ppm	
Ammonia	30		20	NASA 1999[5]
	30	30		US EPA[6] AEGL 1
		20		TRGS 900[7]
Carbon dioxide	13000		13000	NASA 1999
			5000	OSHA (2001)
		5000		TRGS 900
Carbon monoxide	55		20	NASA 1999
	33	27		US EPA AEGL 1
		20		TRGS 900
Ethanol		200		TRGS 900
Methane	5300		5300	NASA 1999
Methanol	30		10	NASA 1999
	530	270		US EPA AEGL 1
		100		TRGS 900

[5] These values **should not be used for situations other than human space flight without careful consideration** of the criteria used to set each value. The SMACs take into account a number of unique factors such as the effect of space-flight stress on human physiology, the uniform good health of astronauts, **and the absence of pregnant or very young individuals**.

[6] U.S. Environmental Protection Agency, AEGL-1 is the airborne concentration of a substance above which it is predicted that the general population, including susceptible individuals, could experience notable discomfort, irritation, or certain asymptomatic non-sensory effects, however, the effects are not disabling and are transient and reversible upon cessation of exposure. AEGL-2 is the airborne concentration of a substance above which it is predicted that the general population, including susceptible individuals, could experience irreversible or other serious, long-lasting adverse health effects or an impaired ability to escape

[7] Technischen Regeln für Gefahrstoffe (TRGS) (German)

Short-term exposure limits are provided by

- NIOSH (2011): 30,000 ppm for 10 minutes
- HSE (2007)[8]: 15,000 ppm workplace
- ACGIH (2005): 30,000 ppm for 15 minutes

It should be noted that according to Harper (2011)[9] the lethal rate is between 1 to 5% for inhaling for 60 minutes air with a CO_2 concentration of 63,000 ppm.

Oxygen concentrations below 15% impair physical and mental performance, and at 10% people lose consciousness. Oxygen deprivation, also known as hypoxia, can have significant effects on the body, ranging from mild to severe. Mild symptoms include dizziness, confusion, and difficulty breathing. More severe cases can lead to coma, loss of consciousness, and even death. Brain damage can occur within minutes of oxygen deprivation, and prolonged hypoxia can cause organ damage, particularly to the heart and brain.

The lethal concentration is 8%.

Temperature

Relevant for the survival of people in distress is their body core temperature. Although a person's ambient temperature influences their body core temperature, this influence also depends on clothing (dry/wet), humidity, wind etc. For instance, a person in wet clothes is on risk for hypothermia at an air of 20°C, while the risk is negligible with dry clothing.

To survive, the human body temperature needs to stay in a certain range, i.e. to avoid death by hypothermia (undercooling) as well as hyperthermia (heatstroke). Hypothermia is defined as the condition when body core temperature is below 35°C and hyperthermia for body core temperature of more than 37.5°C (some literature: 38.3°C). For body core temperature of more than 40°C the term hyperpyrexia is used (many proteins lose their functionality at temperature above 40°C, at temperature above 45°C a process of irreversible denaturation begins). Hypothermia and hyperthermia/hyperpyrexia occur when the body dissipates more heat than it produces, and respectively, produces or absorbs more heat than it can dissipate. It is important to note that hypothermia can occur at temperatures below 40°C.

Therefore, it is recommended to characterise a habitable environment as an environment that allows a person to regulate their body core temperature within the survival range between 35°C and 40°C.

Noise

Noise level between 80 and 85 dB can lead to permanent damage of our hearing (German professional association of ear, nose and throat surgeons) and direct damage is likely for short-term exposure to 120 dB.

Normal communication takes place on noise levels between 40 and 60 dB. Shouting noise level: 70 dB.

Other effects of noise are stress, nervousness, anxiety and concentration disorder.

LSA Code reads

> 4.4.6.9 The lifeboat engine, transmission and engine accessories shall be enclosed in a fire-retardant casing or other suitable arrangements providing similar protection. Such arrangements shall also protect persons from coming into accidental contact with

[8] *EH40/2005 Workplace Exposure Limits*. 1 st. edition.; 2005.
[9] Harper P. Assessment of the major hazard potential of carbon dioxide (CO2). HSE, ed. Published online 2011.

hot or moving parts and protect the engine from exposure to weather and sea. Adequate means shall be provided to reduce the engine noise so that a shouted order can be heard. Starter batteries shall be provided with casings which form a watertight enclosure around the bottom and sides of the batteries. The battery casings shall have a tight fitting top which provides for necessary gas venting.

Space

It is recommended to base the dimensions for seating on anthropometric data.

Considering benches and other seats in which users sit in a row, it is necessary to bear in mind that the breadth of a 95th %ile couple is less than twice that of a 95th %ile individual (S. Pheasant, 2003: Bodyspace. 2nd ed. ISBN 0-7484-0326-4). (The chance of two people, each 95th %ile or more, meeting at random on a bench is only 1 in 400.) In general, n people sitting in a row have a mean breadth of nm and a standard deviation of √n [SD], where m and s are the parameters of the relevant body breadth—which will usually be that of the shoulders.

Example: European elbow to elbow breadth: mean: 478 mm; 95% percentile 540 mm -> difference 62 mm -> SD ~37.7 mm

For 20 persons in a row:

$$breath_{ac} = 20 \cdot 478\,mm + \sqrt{20} \cdot 37.7\,mm = 491.2\,mm$$

Further work is needed to make recommendations on the necessary mobility space, i.e. to assess the impact on health and “will to survive” due to a lack of mobility space.

Provisions

Further work is needed to make recommendations on the necessary provisions (food and water), i.e. to assess the impact on health and “will to survive” due to a lack of provisions.

Annex

References to consolidated functional requirements including addressed hazards

Phase II Proceed to embarkation update III modified SSE 12

Phase III Embarkation update III modified SSE 12

Phase IV Abandon to safe position update III modified SSE 12

Phase Va Waiting for Rescue update III modified SSE 12

Phase Vb Waiting for Rescue update III modified SSE 12

Phase VI Retrieval from SC update III modified SSE 12